

Answers to the reexam in investment and finance, May 2023

Exercise 1 (weight 25%)

Assume that the term structure of interest rates is flat in all scenarios. The table reports the prices of a bond and an option at three different level of interest rates.

Rate (%)	Bond	Option
2.50	99.7573	3.3755
2.70	99.0190	2.9659
2.90	98.4904	2.6471

1. Give a numerical estimate of the sensitivity to an interest rate change of the two assets at a rate of 2.70%.

Answer:

$$\frac{dP_{2.7\%}^{Bond}}{d\lambda} = -316.73 \quad (1)$$

$$\frac{dP_{2.7\%}^{Option}}{d\lambda} = -182.10. \quad (2)$$

2. Assume that you have bought the option and that the rate is 2.70%. Can you perform a local hedge of the option using the bond? (Explain in detail)

Answer:

$$X^{Bond} = -0.5749. \quad (3)$$

3. Assume that the rate decreases by 20 basis points. What is the profit-loss of the hedged position?

Answer:

$$\pi = -0.01488. \quad (4)$$

4. Give an estimate of the modified duration of the two assets when the rate is at 2.70%.

Answer:

$$D_{Mod}^{Bond} = 3.1987 \quad (5)$$

$$D_{Mod}^{Option} = 61.3979. \quad (6)$$

Answer the following questions independently of the first four questions.

5. Compute the price of a perpetuity with semi-annual payments of 10 and yield-to-maturity equal to 5% under semi-annual compounding.

Answer:

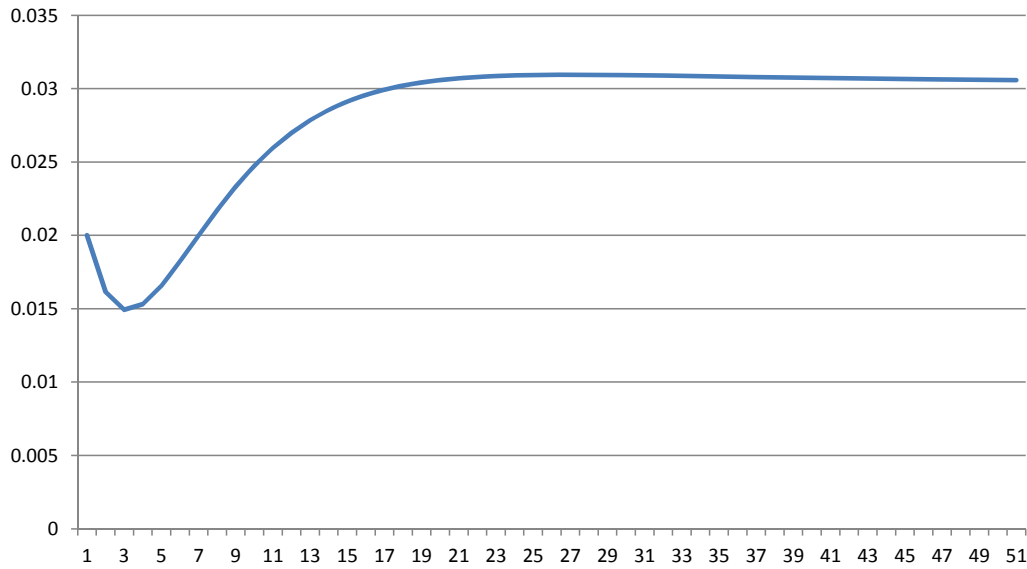
$$P = 400. \quad (7)$$

6. Compute the modified duration of the perpetuity introduced in the previous question.

Answer:

$$D = 20. \quad (8)$$

7. On a given day you observe the depicted spot rate term structure. You want to fit the Nelson-Siegel parametrization to the observed term structure.



$$s_t = \phi_0 + \phi_1 e^{-\frac{t}{\phi_4}} + \frac{\phi_2 t}{\phi_4} e^{-\frac{t}{\phi_4}} + \phi_3 \left(\frac{(1 - e^{-\frac{t}{\phi_4}})}{\frac{t}{\phi_4}} - \left(1 + \frac{1}{2} \frac{t}{\phi_4}\right) e^{-\frac{t}{\phi_4}} \right)$$

Which values would you suggest for the parameters ϕ_0 and ϕ_1 if you had to make a qualified guess without doing any calibrations?

Answer:

$$\phi_0 \approx 0.03 \quad (9)$$

$$\phi_1 \approx -0.01. \quad (10)$$

Exercise 2 (weight 25%)

Consider the stocks X and Y with covariances with the market portfolio $\text{Cov}(r_X, r_M) = 0.01$ and $\text{Cov}(r_Y, r_M) = 0.02$, respectively. The market portfolio has expected return $\bar{r}_M = 0.1625$ and variance $\text{Var}(r_M) = 0.04$. The riskfree rate equals $r_f = 0.095$. Answer the following questions:

1. Compute the CAPM beta and the expected returns of the two stocks according to the CAPM.

Answer:

$$\beta_X = 0.25 \quad (11)$$

$$\beta_Y = 0.5. \quad (12)$$

The two stocks have CAPM returns

$$\bar{r}_X = 0.1119 \quad (13)$$

$$\bar{r}_Y = 0.1288. \quad (14)$$

2. Consider a portfolio with weights in the two assets equal to $w_X = 0.25$ and $w_Y = 0.75$ and with the price 89.50. You assume that the portfolio can be sold for the price of 100 after a year. Should you invest in the portfolio?

Answer: The CAPM expected returns equals

$$\bar{r}_{pf}^{CAPM} = 0.1245,$$

while our private expectation to the expected return equals

$$\bar{r}_{pf}^{Private} = 0.1173 < 0.1245 = \bar{r}_{pf}^{CAPM}. \quad (15)$$

Hence, we should not invest in the asset.

3. What would be your answer if instead the portfolio weights were $w_X = 0.75$ and $w_Y = 0.25$?

Answer: With new weights resulting CAPM expected return

$$\bar{r}_{pf}^{CAPM} = 0.1161 < 0.1173 = \bar{r}_{pf}^{Private}. \quad (16)$$

Hence, in this case we should invest.

4. Assume that the 2-factor APT model holds and that the two factors have risk premiums $\lambda_1 = 0.06$ and $\lambda_2 = 0.08$. Moreover, assume that the portfolio's sensitivity to the two factors are $b_1 = 0.25$ and $b_2 = 0.125$. What is the expected return according to the APT model?

Answer:

$$\bar{r}_{pf}^{APT} = 0.12.$$

5. Give an interpretation of the risk premiums in the APT model.

Answer: Verbal discussion.

6. Determine the portfolio weights that ensure that the CAPM and the APT model are aligned.

Answer:

$$w_X = a = 0.5185, \quad w_Y = (1 - a) = 0.4815. \quad (17)$$

Exercise 3

Consider a state-space model with two periods: Today ($t = 0$) and tomorrow ($t = 1$). The financial market consists of two assets d^1 and d^2 and three possible states $s \in \{1, 2, 3\}$. The possible values of the assets tomorrow are given by the vectors

$$\begin{aligned}(d_1^1 \ d_2^1 \ d_3^1) &= (10 \ 15 \ 22) \\ (d_1^2 \ d_2^2 \ d_3^2) &= (8 \ 17 \ 9)\end{aligned}$$

The prices of the assets today are, respectively, $P_1 = 10.5$ and $P_2 = 8.9$.

1. Argue that the vector $\psi = (0.3 \ 0.25 \ 0.25)$ is a vector of state-prices for this model and that the model is free of arbitrage. Is the market also complete?

Answer: One verifies that ψ is a vector of state-prices by checking that

$$\begin{pmatrix} 10 & 15 & 22 \\ 8 & 17 & 9 \end{pmatrix} \begin{pmatrix} 0.3 \\ 0.25 \\ 0.25 \end{pmatrix} = (10.5 \ 8.9)$$

As the state-prices are strictly positive, the economy is free of arbitrage. The economy is however incomplete, as we have fewer assets than states.

2. What is the risk-free rate implied by the state prices ψ ?

Answer:

$$r_f = \frac{1}{\psi_0} = \frac{1}{0.3 + 0.25 + 0.25} - 1 = 0.25$$

3. Assume that a call option on asset 1 with strike $K = 12$ is traded in the market. What price would you quote for this call option?

Answer:

$$C_0 = 4.0625$$

4. A third asset d^3 is introduced to the market, with possible payments

$$d^3 = (5 \ 15 \ 20)$$

what is the price of this asset, P_3 , given state prices ψ ?

Answer:

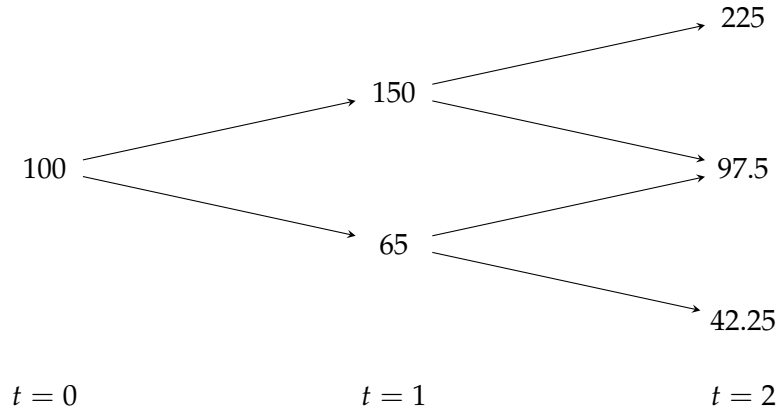
$$P_3 = 10.25$$

5. It turns out that the price of asset 3 is observed in the market as $P_3 = 7$. Is this consistent with no arbitrage?

Answer: Yes.

Exercise 4

Consider the following binomial lattice model for the price evolution of a non-dividend paying stock S over two periods:



The constant, discretely compounded risk-free rate is $r = 5\%$.

1. What is the risk-neutral probability of ending up in state $S(2) = 97.5$?

Answer:

$$Q(S(2) = 97.5) = 0.4983$$

2. What is the price of an American call and put option on S with strike $K = 85$ and maturity $T = 2$?

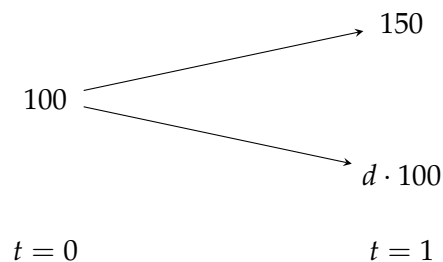
Answer:

$$C_0^a = 33.7718$$

As for the American put, we have to consider whether it is beneficial to exercise early. We have

$$P_0^a = 10.8674$$

Assume now that the first period, the price evolves as



where $d \geq 0$.

3. Determine the interval $[\underline{d}, \bar{d}]$ with $\underline{d} \leq d \leq \bar{d}$ such that the model is free of arbitrage.

Answer: $[\underline{d}, \bar{d}] = [0, 1.05]$.

4. Determine the possible interval of European call prices $[\underline{C}, \overline{C}]$ with strike $K = 85$ given that $d \in [\underline{d}, \overline{d}]$.

Answer:

$$\underline{C} = 22.9092, \quad \overline{C} = 62.2222$$

Assume now that the stock price follows a geometric Brownian motion:

$$dS(t) = \mu S(t)dt + \sigma S(t)dW_t$$

where W_t is a Brownian motion and $\mu, \sigma > 0$. The interest rate r is still constant, but now *continuously* compounded.

5. A European call option with strike $K = 100$ and maturity $T = 1.25$ years is trading with an implied volatility of 0.35. What is the price of this option?

Answer:

$$C = 18.2604$$

6. There is also a derivative on the market with payoff function $\sqrt{S(T)}$. You are told that the pricing function F of this derivative is given as

$$F_t = \sqrt{S(t)} e^{-\frac{1}{2}(r+\sigma^2)(T-t)}$$

is this pricing function consistent with no arbitrage?

Answer:

$$\frac{\partial F}{\partial t} + rS \frac{\partial F}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 F}{\partial S^2} = -\frac{1}{2}rF_t \neq 0$$

So this price process is not consistent with no-arbitrage.